Network Analysis of Character Interactions in J.R.R. Tolkien's The Hobbit

2. Abstract

In this study, we analyzed the structure of the character interactions in *The Hobbit* by J.R.R Tolkien. The dataset that was used represents the character interactions network given from the novel. The interactions are shown as nodes, while the interactions/co-appearances are represented as edges. The network is undirected and unweighted. It contains 41 nodes and 160 edges. Additionally, there is a density of 0.1951 and one connected component.

In order to analyze this dataset, we used several network methods using Python and the Jupyter Notebook environment. Our methods include the calculation of key network metrics, including degree centrality, closeness centrality, and betweenness centrality. Additionally, the structural analyses included k-core decomposition and clique detection. With these techniques, it allows for the identification of influential characters and the examination of the organization of the narrative network.

The results show that Bilbo Baggins has the highest degree centrality. This indicates that he is interacting with the largest number of characters in *The Hobbit*. Additionally, Thorin Oakenshield was shown to have the highest betweenness centrality. Which demonstrates his role as a bridge connecting different character groups. Overall, these findings illustrate how network analysis provides many quantitative insights into the narrative structure, as well as emphasize the significance of important key characters in maintaining the cohesion of the story.

3. Literature Review

Network science is now being used in many fields to analyze complex systems. For example, its use is increasing in various fields including sociology, physics, biology, and digital humanities [**7**] [**15**]. Network analysis is also used in literary research as a computational method, through which the relationships between the characters of a story or novel can be analyzed [**4**] [**5**]. It is possible to show a literary story or novel through a graph. Here, each character is considered as a node and the relationship or communication between the characters is shown as an edge [**4**] [**17**]. As a result, researchers can easily analyze the structure of the story and the relationships between the characters using various measures or methods of graph theory.Through this method, it is easy to understand which characters are more important in the story, which characters are more connected to others, and how their relationships advance the story. Therefore, character network analysis is an effective and important method for understanding the structure of literary writing and the social relationships of characters [**3**] [**5**].

An important concept in this case is the character network, which represents the social structure of a story. In such a network, each character is considered a node and the connections or relationships between characters are shown as edges [**4**] [**17**]. Labatut and Bost (2019) have shown that using character networks, researchers can analyze literary stories through computational methods. As a result, many relationships and patterns of interaction can be seen, which are not easily understood by simply analyzing the text [4]. By extracting the relationships between characters from literary texts and showing them in the form of graphs, researchers can understand how similar or different the story network is to real-life social networks [**5**].

Many previous studies have shown that the structure of many literary networks is similar to real-life social networks. For example, MacCarron and Kenna analyzed mythological stories and showed that the communication networks of characters in epics such as The Iliad resemble real social networks in many ways [**6**]. For example, there is clustering and hierarchical organization. However, they also noted that some mythological stories contain exaggerated or completely fictional elements. This makes the structure of the network somewhat unusual. If some highly connected characters are changed or removed, the structure of the network resembles real social networks more closely. These results suggest that it is possible to use network analysis to understand the differences between real and fictional social structures within a story.

Another important feature of character networks is the small-world phenomenon. This means that a network has many clusters and the distances between nodes are relatively small. Stiller, Nettle, and Dunbar (2003) analyzed networks from Shakespeare's plays and showed that the interactions of characters often create a small-world structure [**8**]. In such networks, characters are connected to each other by relatively short paths. As a result, information or influence can spread very quickly throughout the network. The presence of small-world features in literary stories shows that writers often create relationships between characters in a way that closely resembles real-life human communication networks [**8**] [**19**]. This type of structure keeps the story well-organized while simultaneously incorporating many characters and various subplots into a relatively small social structure.

Centrality measures play a very important role in network analysis. They are used to understand which nodes in a network are the most influential or important [**9**]. In literary networks, the centrality metric is used to understand which characters are in the most important position in the structure of the story [**12**]. Degree centrality shows how many other characters a character has direct connections to. This shows how socially active that character is in the story. Betweenness centrality identifies characters who create connections between different groups of characters. That is, they work to connect different parts of the story together. Eigenvector centrality determines the influence of a character based on the importance of other characters associated with it. That is, the more connected that character is to important characters, the more influential that character is. Using these measures, researchers can identify the main characters, important intermediary characters, and supporting characters in a story. This type of analysis can help us understand how characters interact and influence each other within a story.

In addition to identifying dominant characters, network analysis also considers the stability and robustness of the network. Robustness refers to the extent to which a network can retain its original structure if some nodes or edges are removed from the network [**15**]. Analyzing robustness in literary networks can help us understand how much the structure of a story depends on certain characters [**11**]. Research has shown that networks in which some nodes are highly connected tend to be quite stable even after sudden or random changes. However, if very important or central nodes are removed, the structure of the network can become weak [**10**]. This feature is also important in story structure. Because if a central character in a story is removed, the connectivity of the entire network can change significantly and the structure of the story can also change significantly. Therefore, analysing network robustness provides insights into how narratives rely on key characters to maintain their structural coherence.

Another structural feature frequently observed in complex networks is the core–periphery structure. In such networks, a few highly connected nodes usually form a dense group, called the core. On the other hand, nodes that are relatively less connected are located at the periphery or edge of the network. In literary stories, core characters are usually the main characters or important figures. They interact with each other more and move the story forward. Peripheral characters, on the other hand, are less connected and are usually seen in certain scenes or anecdotes. By identifying the core-periphery structure in character networks, researchers can understand how the story structure is organized hierarchically and what roles different characters play in the story [**18**].

Franco Moretti further expanded the use of network theory in literary analysis. He proposed a network-based approach to understanding the plot of a story. According to Moretti (2011), traditional literary criticism usually places more emphasis on the interpretation of individual characters or themes. However, using network analysis, researchers can analyze the relationships between characters in the context of the whole system [**7**]. By visually showing the characters' communication networks, researchers can find different clusters, identify smaller subgroups, and analyze the structural organization of the entire story. This approach allows literary scholars to study narrative systems quantitatively while still maintaining connections to qualitative interpretation.

Although network analysis is a powerful method for analyzing stories or narratives, it has some limitations. Network models usually present stories in a simplified way and focus mainly on the relationships between characters [**4**] [**14**]. As a result, other important aspects of literature, such as symbolism, themes, and writing style, can often be overlooked. In addition, the process of extracting character relationships from text can also be problematic. Especially when this task is done manually, there can be personal interpretations or inconsistencies [**5**]. Nevertheless, many researchers believe that network analysis supports traditional literary analysis methods and provides a quantitative idea of the structure of the story [**2**].

The dataset used in this study is a character interaction network created from J.R.R. Tolkien's novel The Hobbit. Here, each character is considered a node and the connections or relationships between characters are shown as edges [**11**]. In this way, a social network is created that represents the relationships within the story. It is possible to analyze the structure of this character network using various methods of network analysis, such as centrality measures and structural analysis [**9**] [**12**]. Through this, important or influential characters in the story can be identified. Building on the ideas discussed in previous studies, this study aims to analyze the network structure of The Hobbit dataset. Specifically, it will analyze centrality, connectivity, robustness, and whether there is a core-periphery structure in the character network [**18**].

4. Dataset Description

What is the dataset?

The dataset that was used in this analysis is a social network representing the interactions between characters in J.R.R. Tolkien’s novel, The Hobbit. The data captures characters' co-appearance or interaction of characters throughout the story, allowing us to map the social structure and network of the Hobbit's world.

Network Type

The Hobbit network is an undirected and unweighted network. It is undirected because an interaction between two characters is considered a mutual event; for example, if Bilbo interacts with Gandalf, the connection exists equally for both nodes. Moreover, the edges are unweighted, meaning we are focusing only on the existence of a connection, not the frequency of their meeting.

Network Elements

- **Nodes:** Represent individual characters appearing in the story.
- **Edges:** Represent interactions. If two characters appear together or interact with each other, an edge is created between them.

Network Statistics

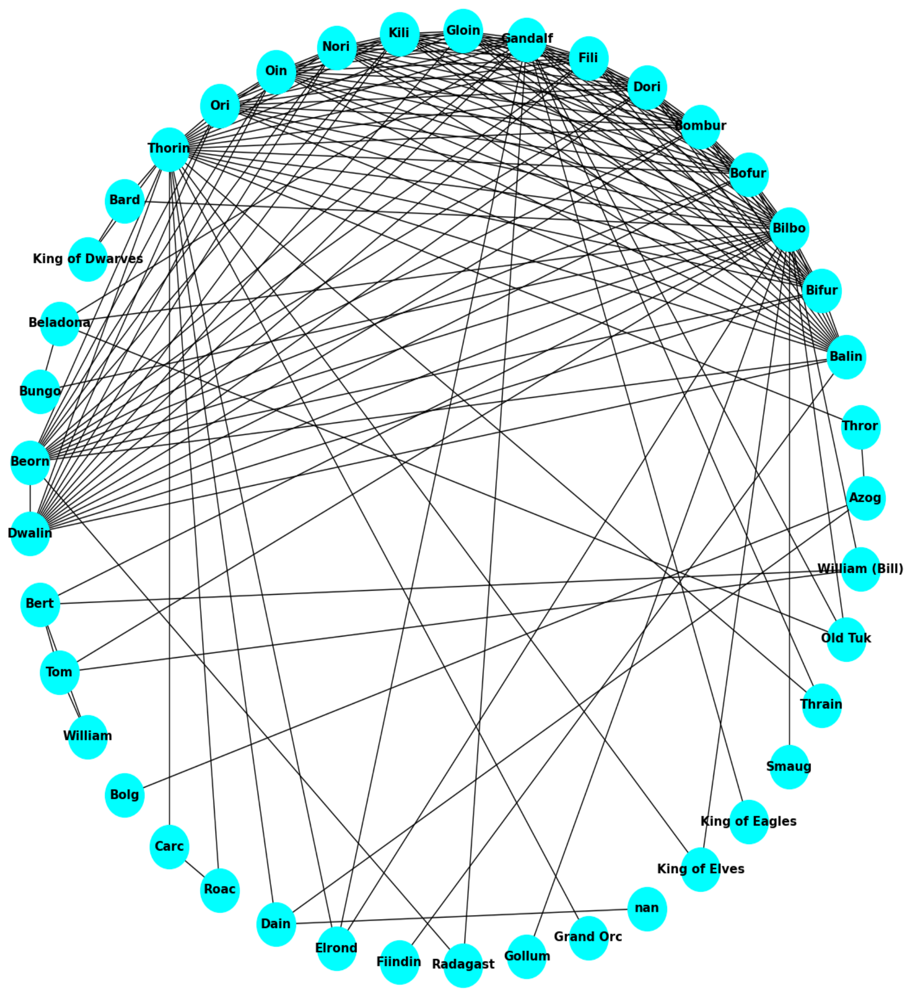
In this analysis, we are using the NetworkX library in Python; we calculated the following metrics for our network (G):

- **Number of Nodes**: 41
- **Number of Edges**: 160
- **Network Density**: 0.1951 or 19.51%
- **Mean Degree**: 7.8
- **Component**: 1

5. Network Analysis

This section presents the analysis of the network (mathematical properties, visualization, algorithm computation, and statistical methods). It covers the dataset used, the preprocessing step of the dataset, the environment the data was analyzed in, and the measures used, as well as findings obtained from the analysis.

The dataset used in this analysis has been described in section 4. Above, the network contains **41 nodes** and **160 edges**, which is an indication that the network is a moderately connected structure. The density of this network is relatively low, and this suggests that not all the characters interact frequently with each other in the network. The graph of the network is shown below



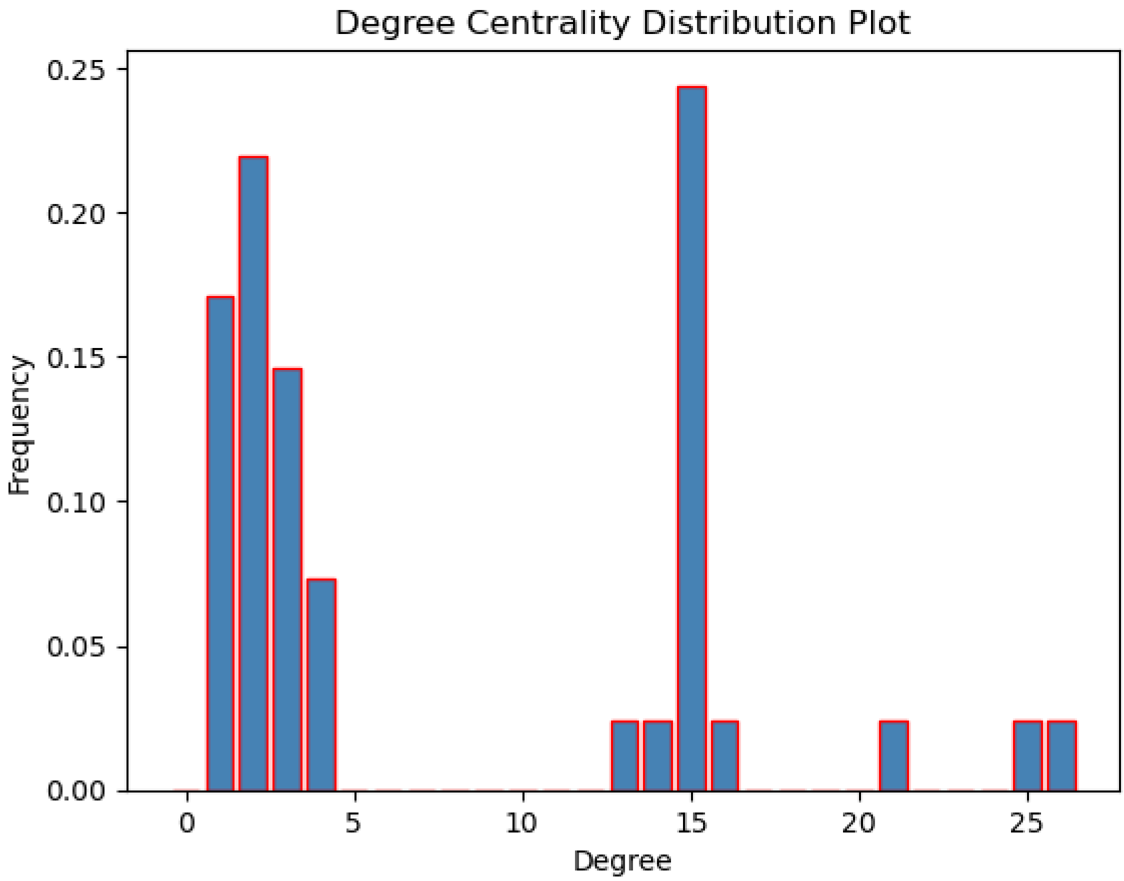
Mathematical Properties (Network Metrics) Interpretation

Density: It is defined to be the ratio of the real number of edges with respect to the maximum possible number of edges. In this network, we got our density as **0.195,** and the possible number of edges in this network is 820.

Degree Centrality: nodes with high degree centrality often represent characters that interact with many others in the story; these characters are most likely to be focused on in the narrative, as so many other characters are connected to them in the network. In our dataset, we got a few characters with notably high degree centrality; they are presented below: *"**Bilbo with 26 degrees," "Thorin with 25 degrees," and "Gandalf with 21 degrees.**"*

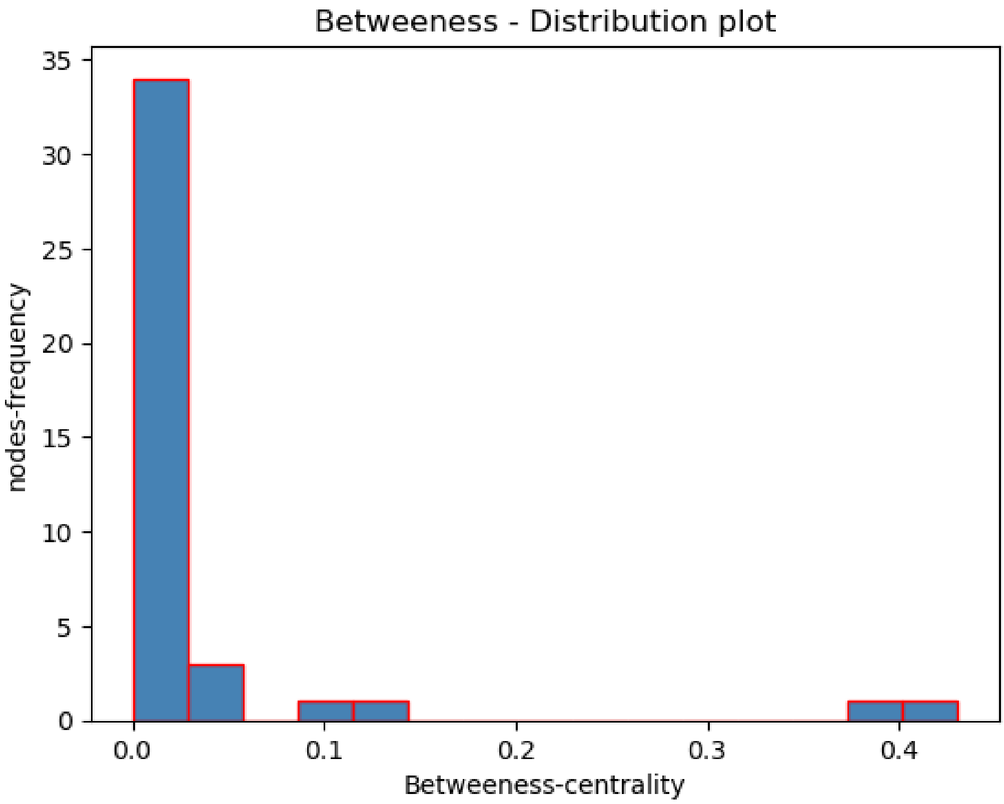
The average degree of the networks is calculated as **7.80**. This result is interpreted to mean that the average character in the network interacts fairly with at least 8 other individuals, so we can classify any character with a degree above **7.8** as a main character in this network.

The degree distribution table presented is in the appendix, and the distribution plot is shown below; it shows that most nodes have a relatively low value, and only a few have high degree values. These tell us that the majority of nodes interact with only a small number of others in the network.



__Betweenness Centrality__: It measures how often a node lies on the shortest path between other nodes (characters); nodes with high betweenness act as bridges connecting different parts of the network. The role of these nodes is so important that without their existence we may end up having separate groups of characters in the network. Nodes with low betweenness in the network are likely to be peripheral characters, and they play a minor role in connecting different groups.In our analysis, we discovered the node (character) with the maximum betweenness as ***Thorin*** *with a betweenness value of* ***0.43*** *and also the nodes (characters) with the minimum betweenness of* ***0.0****, which are the likes of* ***King_of_elves, William, etc.***

The distribution plot below shows the betweenness centrality in the network

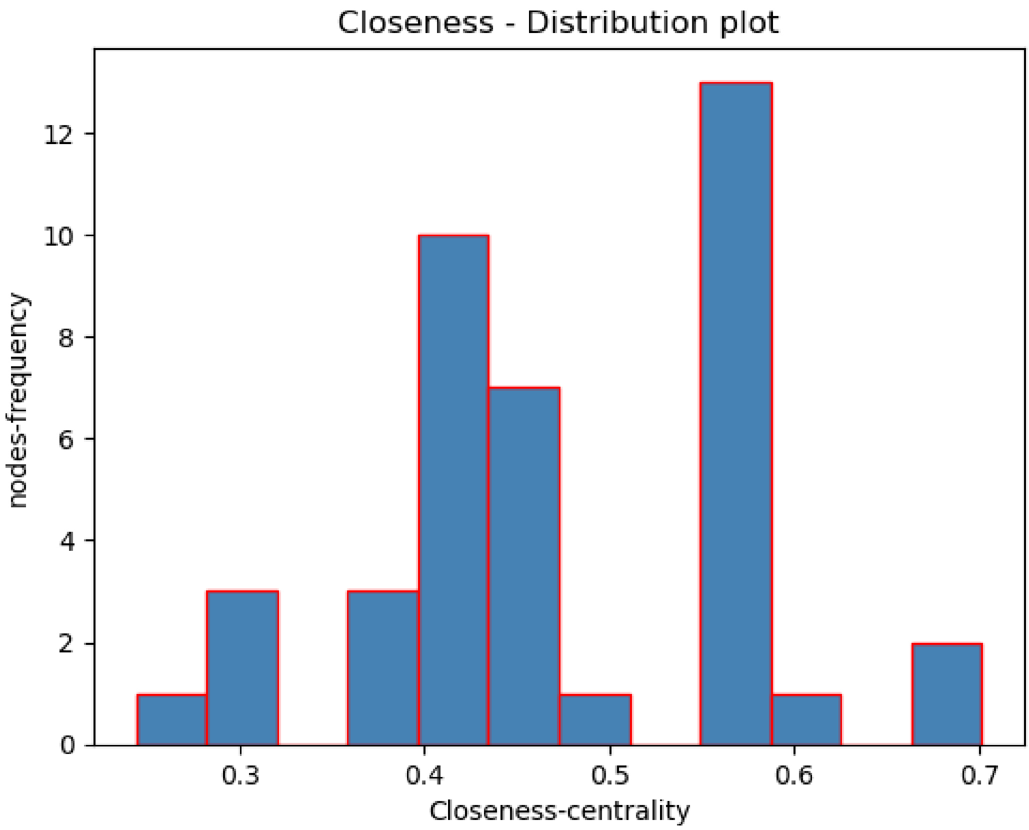


__Closeness Centrality__: It measures how close to a particular node other nodes are in the network. The nodes (characters) with high closeness centrality are positioned in the center so they can interact with others.

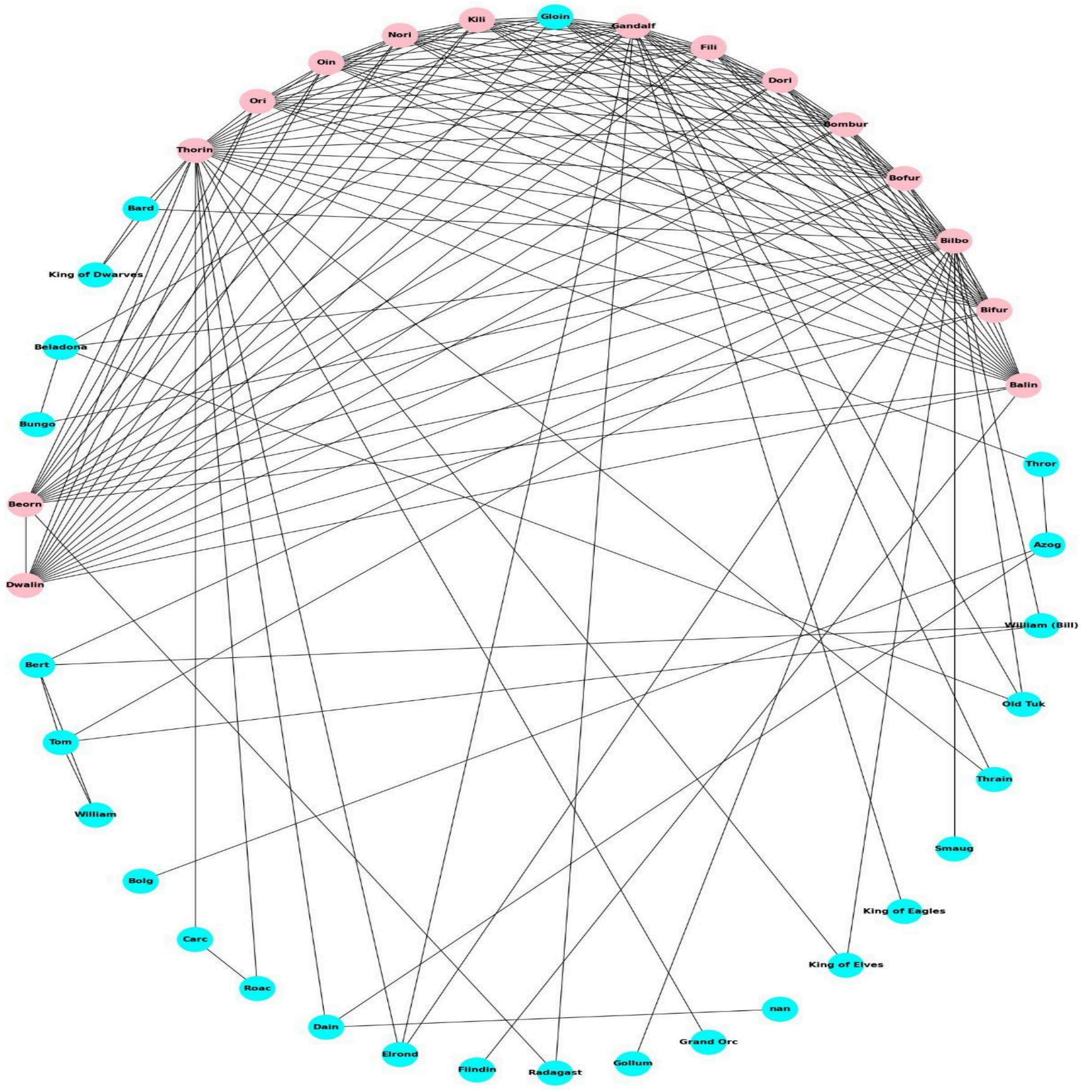
The mean closeness of the network is **0.447,** which indicates that, on average, each node is moderately close to the rest of the nodes in the network. This suggests that the interaction in the network is being spread efficiently between nodes in the network.

Nodes with high closeness values in the network are **Thorin and Bilbo**; it means these particular characters can reach every other character quickly compared to nodes with lower closeness.

The distribution plot below shows the closeness centrality in the network



K-core: It helps us identify the least, evenly, or most densely connected group in the network.
Nodes belonging to a subset with high core values show that they are fully connected to each other, and they represent the most structurally important character in this network.
The highest core value in this network is calculated as **14,** and we have few nodes belonging to this core.
15 nodes belonging to k-14 core (maximum core): 'Oin,' 'Beorn,' 'Dori,' 'Bombur,' 'Ori,' 'Bifur,' 'Balin,' 'Fili,' 'Bilbo,' 'Kili,' 'Nori,' 'Dwalin,' 'Bofur,' 'Gandalf,' and 'Thorin.' The graph of their connectivity is shown below



K-component: A k-component is a set of nodes such that each is reachable from each of the others by at least k-node independent paths. From our analysis we got a maximum k-component of **1.** This indicates that each set of nodes in our network is reachable from at least **1-node** independent path.

Cliques: In our network, cliques represent characters that collaborate together. Cliques with larger nodes represent groups that are highly interconnected and vice versa.
From our analysis, we grouped clique by sizes; they are listed below

Size 15 cliques: 1 cliques
Size 14 cliques: 1 cliques
Size 4 cliques: 3 cliques

Size 3 cliques: 8 cliques
Size 2 cliques: 11 cliques
The number of clique with maximum size here is **1**

6. Discussion

6.1 Network Implications (Density and Cohesion)

Density Analysis: The network has a density of 0.1951, or 19.51%, which tells us that there is a sparse social structure in the Hobbit's network. In the context of The Hobbit, this tells us that the narrative is not a community story where everyone knows everyone and not every character interacts with each other, but it's a linear journey where characters are met in some chapter or some part of the story and then rarely interact again. For example, Bilbo meets Gollum once, but they never interact again, which is a pattern reflected in the low density score.

Network Cohesion: Despite this sparsity, the network is cohesive, proved by the result of having only 1 connected component. Meaning that every character, no matter how briefly they appear, will be linked back to the main story in some way, this demonstrating Tolkien's great narrative story building.

Key Structural Nodes: Our analysis identifies Bilbo and Thorin as the pillars of the network. While Bilbo acts as the primary social hub holding the highest Degree Centrality of 26, Thorin's role is the bridge that provides structural reach to the others group or faction, as seen in his leading Betweenness Centrality of 0.431.

6.2 Robustness: Removing Central Characters

- **Impact of removing Bridge character** : If we were to remove Thorin, who holds the highest Betweenness Centrality of 0.431, the network would likely break into isolated clusters.
- **Narrative Impact**: Because Thorin sits on over 43% of the shortest paths between character pairs, his removal would break the connections between various groups and factions. Certain factions in the story would never have met as they did in the original narrative, which will impact the narrative of the story.
- **The Structural Glue**: This mathematically confirms that while Bilbo is the protagonist and social hub, Thorin is the structural glue. He makes interactions between groups possible and holds together the characters that drive the narrative forward. Without Thorin's node, the social web of *The Hobbit* would have collapsed into disconnected smaller groups.

6.3 Comparison with Literature

- **Small-World Properties**: Our findings are consistent with the research of Mac Carron and Kenna (2012), who argued that Tolkien's literary universes exhibit "Small World" characteristics [6]. While this study focused on global metrics, our result of a low overall density (0.1951) with a single connected component aligns with their theory that Tolkien's worlds are vast but reachable through key character connections.

- **The Protagonist-Bridge Duality:** Similar to the methodology used by Beveridge and Shan (2016) in their study of Game of Thrones, our data highlights a clear distinction between the protagonist and the connector character [1]. While Bilbo's role is the primary social hub with the highest degree centrality of 26, interacting with more than half of the characters in the story, he is primarily a measure of character visibility. In contrast, Thorin acts as the "connector" or "bridge" with a superior Betweenness Centrality of 0.431. This confirms that in complex narratives, the character with the most interactions is not necessarily the one providing the most structural stability to the plot. It also demonstrates that the protagonist is not the only force keeping the story moving forward.

6.4 Limitations and Further Research

Limitations:

- The study uses an unweighted network, meaning every interaction is treated as equal; no frequency of interaction is involved. In reality, some characters have deep, frequent relationships, while others only meet once; however, the current model does not account for this weight or frequency of interaction.
- The dataset represents a static snapshot rather than the full evolution of the narrative, meaning it does not capture how the social network grows and shifts as the journey progresses from chapter to chapter.

Further Research:

- Furthermore, adding edge weights and transitioning to a directed network dataset would provide a much more precise and detailed view of character closeness and influence. Such an approach would allow for a more nuanced analysis of which characters share the strongest bonds versus those with weak or transient relationships, ultimately offering a more complete picture of the social dynamics within the text.

7. Conclusion

In conclusion, this project has successfully applied a network analysis method to quantify the social architecture of J.R.R. Tolkien's *The Hobbit*. By transforming character interactions into a graph of 41 nodes and 160 edges, we identified two distinct types of character importance that drive the narrative forward. While Bilbo Baggins serves as the primary social hub with a degree centrality of 26, reflecting his role as the protagonist, Thorin Oakenshield functions as the critical structural holder and connector with a leading betweenness centrality of 0.431.

These findings matter because they demonstrate that a character's importance is multi-dimensional, meaning visibility of protagonists does not always equal structural power.

The network’s low density (19.51%) and single connected component are mathematically consistent with small world properties, proving that the story is a cohesive but linear journey through a vast and sparse world. Ultimately, this study shows that network analysis is a powerful method for understanding literary works, revealing the detailed character interactions that support the plot and drive the narrative progression. Furthermore, this analysis can serve as a valuable case study or guideline for writers and literary researchers in the future.

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Dataset source

https://github.com/aholanda/charnet/blob/master/data/hobbit.dat

https://icon.colorado.edu/networks

9. Appendix

The complete implementation of the network analysis, including the computation of centrality measures, components, and cliques, was carried out using Python in a Jupyter Notebook. The full notebook can be accessed using the link below.

Jupyter Notebook Code